FPT SCHOOL OF BUSINESS & TECHNOLOGY (FSB)

Course: Machine Learning Engineering (MLE501) • Class: MSA41HCM (Fall 2026)

Consumer Credit Scoring from Multi-Source Data

An Explainable Logistic Regression Scorecard and Source-Level Ablation Study

Trần Tiến Cường
Student ID / Email: cuong26ms23348@fsb.edu.vn

EXECUTIVE ABSTRACT

TNF Finance operates in point-of-sale (POS) consumer installment lending across roughly 300 dealerships in southern Vietnam. The portfolio historically experienced a high default rate of 11.86%, burdened by slow and inconsistent manual underwriting. This study integrates six fragmented internal and external databases covering 20,000 loan applications while adhering strictly to Point-in-Time validation. An explainable Logistic Regression scorecard is established, yielding a ROC-AUC of 0.8262 and a Kolmogorov-Smirnov (KS) statistic of 0.4972 under Stratified 5-Fold Cross-Validation. Incorporating Credit Bureau records produced the largest single performance leap (+9.23% AUC), while cross-source metrics (Total Debt-to-Income and Income Exaggeration) provided critical resistance against application fraud. An operational cut-off threshold of 0.15 is proposed, enabling a 76.6% automated approval rate while more than halving post-approval bad debt to 5.34%.

1. INTRODUCTION & BUSINESS PROBLEM

In retail installment financing (motorbikes, consumer electronics, and mobile phones), underwriting speed at the point of sale is a critical competitive advantage. However, retail loan applications inherently rely on self-declared customer submissions. Crucial determinants of actual creditworthiness—such as unverified earnings, external debt liabilities, and historical repayment discipline—are distributed across disparate legacy systems.

The objective of this assignment is to develop an automated, fully explainable credit scoring engine utilizing Logistic Regression to estimate the default probability of each incoming application upon submission.

2. PROBLEM FORMULATION & POINT-IN-TIME PRINCIPLE

2.1. Analytical Unit and Target Label

- **Unit of Observation:** A single loan application (`ma_ho_so`) captured at the time of store submission.

- **Target Variable (`vo_no`):** Defined as `vo_no = 1` if the borrower incurs 90 or more days past due (DPD ≥ 90) within the first 12 months on book, and `0` otherwise.
- **Dataset Scope:** 16,046 labeled training observations (1,903 defaults, baseline bad rate of 11.86%) and 3,954 test records with withheld labels.

2.2. Strict Point-in-Time Information Principle

To prevent look-ahead bias and data leakage, all features must be computed strictly using information known on or prior to the application filing date (`ngay_nop`):

- **Historical Repayment Records (`repayment_history.csv`):** Contains 179,265 historical payment periods. All scheduled dues concluded prior to December 1, 2023, while all incoming applications commenced in January 2024. Consequently, this source poses zero risk of temporal leakage.
- **Payroll Statements (`payroll.csv`):** Covers 259,416 monthly payroll entries from July 2023 to June 2025, operating concurrently with applications. *Temporal Guard:* For each applicant, records were filtered strictly to `thang ≤ thang(ngay_nop)`. Any earnings recorded subsequent to the application month were systematically excluded.

3. MULTI-SOURCE DATA INTEGRATION

The data model consolidates six disparate operational sources into a unified analytical base table maintaining exactly 20,000 application rows (16,046 Train and 3,954 Test):

Table 1: Data Integration Schema and Aggregation Logic

Source Table	Join Key	Cardinality	Extracted & Aggregated Features
`customers.csv`	`ma_khach_hang`	N : 1	Demographic attributes, age, marital status, residential tenure, housing type.
`dealers.csv`	`ma_dai_ly`	N : 1	Dealership scale (small, medium, large), product specialization, partnership tenure.
`credit_bureau.csv`	`ma_ho_so`	1 : N → Agg	Total outstanding external balance, monthly repayment sum, max risk group (1–5), 12-month delinquency counts.
`repayment_history.csv`	`ma_khach_hang`	1 : N → Agg	Historical maximum DPD, mean positive DPD, count of late payments, underpayment frequency.
`payroll.csv`	`ma_khach_hang`	1 : N → Agg (Point-in-Time)	Filtered by application date: Latest verified salary, historical mean salary, earnings volatility (standard deviation).

Treatment of Missing Segments:

- **Absence of Bureau Records (38.86%):** Indicates no active debt at other registered institutions. Balances and overdue counts were assigned zero, alongside an indicator flag `has_bureau = 0` (default rate: 5.79%).
- **First-Time Borrowers (59.92%):** Applicants with no previous borrowing history at TNF were assigned zero for historical delinquency, identified via `is_existing_cust = 0`.
- **Informal Sector Workers (45.83%):** Applicants lacking employer payroll linkages were flagged with `has_payroll = 0`, utilizing self-declared income as the proxy baseline.

4. CROSS-SOURCE FEATURE ENGINEERING

The most potent predictors emerged from cross-referencing self-declarations against verified external records:

4.1. Total Debt-to-Income (Total DTI)

Total DTI = (New TNF Monthly Payment + External Bureau Monthly Debt) / Estimated Monthly Income

Empirical Finding: Applicants in the highest DTI quartile (> 37%) exhibited a default rate of **23.01%**, compared to just **5.18%** in the lowest quartile (≤ 14.3%). Total DTI captures aggregate household leverage that siloed appraisals routinely miss.

4.2. Income Exaggeration Factor

Income Exaggeration = 1 - (Latest Verified Payroll Salary / Self-Declared Stated Income)

Empirical Finding: For applicants with verified payroll, exaggerating declared earnings by over 33% resulted in a default surge to **22.32%** (versus 5.66% among accurate disclosers). This factor unmasks moral hazard during loan origination.

4.3. Equity and Delinquency Features

- **Down-Payment Ratio (`tra_truoc / gia_tri_hang`):** Reflects customer commitment; higher equity significantly lowers default risk.
- **Historical Delinquency (`repay_dpd_max`):** Past payment tardiness at TNF remains a durable indicator of future recidivism.

5. MODEL TRAINING & ABLATION STUDY

An L_2-regularized Logistic Regression model (C = 0.1, RobustScaler for continuous variables, One-Hot Encoding for categorical factors) was evaluated across five cumulative data tiers via **Stratified 5-Fold Cross-Validation**.

Table 2: Ablation Study Results across Five Cumulative Data Stages

Data Stage	Features	ROC-AUC (5-Fold)	KS Statistic	ΔAUC	Key Takeaway
1. Application Only	9	0.6567	0.2334	—	Self-reported information yields weak standalone separation.
2. + Customer & Dealer	20	0.6838	0.2686	+0.0271	Adds residential stability and distribution channel quality.
3. + Credit Bureau (CIC)	26	**0.7761**	**0.4131**	**+0.0923**	**Largest single leap (+9.23%).** Vital visibility on outside leverage.
4. + TNF Repayment History	34	0.8003	0.4510	+0.0242	Internal track record meets banking industry standards (AUC ≥ 0.80).
5. + Payroll & Cross-Features	41	**0.8262**	**0.4972**	+0.0259	**Optimal model.** Directly mitigates income fraud and excessive DTI.

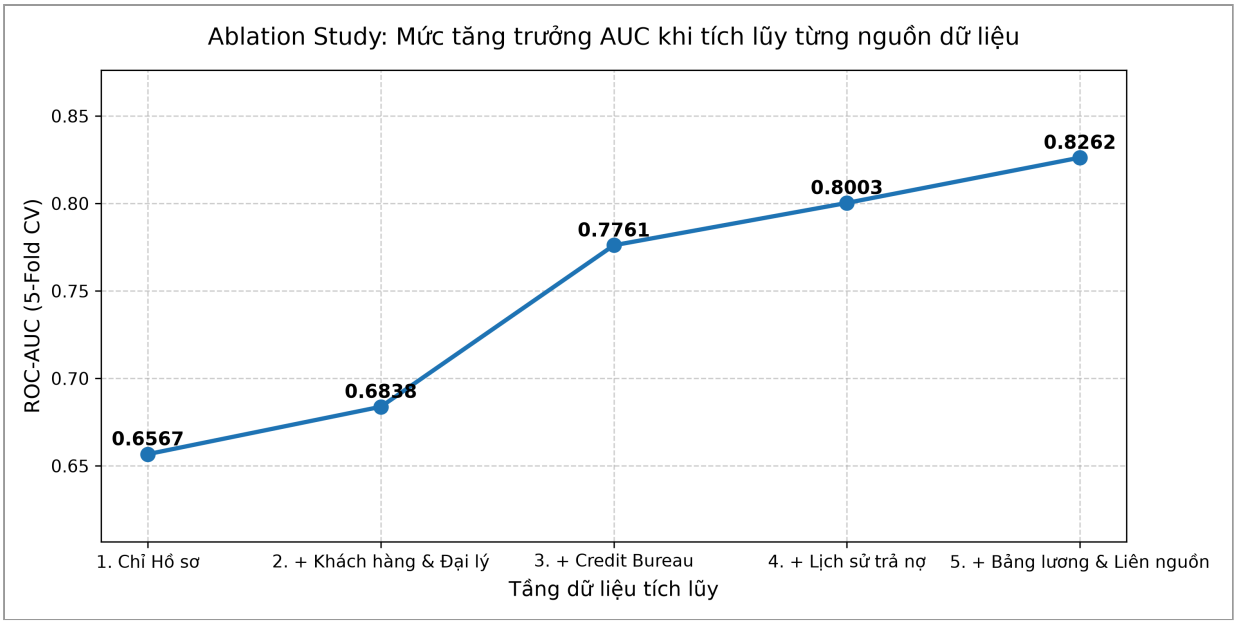


Figure 1: Progressive ROC-AUC improvement across cumulative data tiers.

Data ROI Assessment: Credit Bureau inquiry generates the highest marginal economic return, driving AUC past 0.77 and elevating the KS statistic above the institutional benchmark of 40%. Payroll statements and cross-source variables provide the second critical lift, closing underwriting vulnerabilities against stated income inflation.

6. OPERATIONAL CUT-OFF STRATEGY

In live underwriting, the model outputs estimated default probability $P \in [0, 1]$. Determining an operational cut-off reflects the structural trade-off between loan acquisition volume and loss containment:

Table 3: Confusion Matrix and Operational Trade-Offs by Cut-Off Threshold

Cut-Off Threshold	Approval Rate	Post-Approval Bad Rate	Defaulters Caught (Recall)	Strategic Policy Posture
0.10	65.6%	3.96%	78.1% (1,486 / 1,903)	Defensive posture for contractionary economic conditions.
0.15 (Recommended)	**76.6%**	**5.34%**	**65.5% (1,246 / 1,903)**	**Balanced baseline:** Approves ~3/4 of leads while halving bad rate (11.86% → 5.34%).
0.20	83.2%	6.36%	55.4% (1,054 / 1,903)	Expansionary posture for promotional seasonal campaigns.
0.30	90.3%	7.68%	41.5% (790 / 1,903)	Lenient policy, eliminating only extreme tail-risk defaulters.

Policy Recommendation: Implement **Threshold 0.15** as standard operating policy. Applicants with $P < 0.15$ qualify for Straight-Through Processing (STP) in under 30 seconds at partner checkouts. Applicants with $P \geq 0.15$ are rejected or routed for manual review requiring supplementary down-payments or co-signers.

7. PRACTICAL INSIGHTS, LIMITATIONS & MAINTENANCE

7.1. Empirical Risk Findings

- **Repeat Borrowers Are Not Inherently Safer:** Existing TNF clients demonstrated a higher aggregate default rate (13.23% vs. 10.94% for new borrowers). While pristine historical clients (DPD = 0) represent prime profiles, repeat borrowers who struggled in prior contracts frequently refinance to roll over distressed debt. Tenure should never warrant automatic approval.
- **Product-Specific Risk Heterogeneity:** Motorbike financing exhibits the highest default rate (13.58%) due to larger ticket sizes and long commitments (18–24 months). Conversely, mobile phones demonstrate the lowest delinquency (10.70%) due to shorter durations and irreplaceable utility in daily livelihoods.

7.2. Limitations & Model Governance

- **Informal Sector Underwriting:** Approximately 46% of retail borrowers lack formal payroll verification. TNF should explore alternative data sources (telecom credit scoring and utility history) to supplement income estimation.
- **Stability Monitoring:** Track the Population Stability Index (PSI) monthly on critical features (`dti_tong`, `bureau_tong_du_no`). If PSI exceeds 0.25 (indicating distribution drift), model recalibration should be scheduled immediately.

8. SUBMISSION DELIVERABLES

- **Pipeline Notebook:** `notebooks/credit_scoring_pipeline.ipynb` — Fully executable Python notebook covering data ingestion, cross-validation, and ablation analysis.
- **Predictions File:** `outputs/du_doan.csv` — Exact 3,954 predictions for the hidden test evaluation.
- **Report Document:** `report/report.html` — Complete academic project report.